collect_stats.py

`scripts/collect_stats.py · 107 lines`

What it does

Runs the calibration data through the original model and accumulates EASY-EP statistics for every MoE layer (both stacks for encoder-decoder models), excluding padding.

Run

```
uv run python scripts/collect_stats.py
```

Pipeline order: inspect_model -> build_c4_support_eval -> collect_stats -> score_experts -> replace_experts -> evaluate (before_training) -> train_biases -> evaluate (after_training) -> export_model

Reads

- the original model
- `CALIBRATION_DATASET_PATH`

Writes

- `<stats_dir>/expert_stats.pt` (running sums)
- `expert_stats_manifest.json` (calibration fingerprint, layers, runtime, memory)

Notes

- Checks gate consistency and finite outputs on every batch.

Defined in this file

Extracted from the source code (classes, public functions and methods).

Name	Description
`main()`	Script entry point: runs this stage end to end.

Settings it reads

`CALIBRATION_BATCH_SIZE`, `MODEL_DTYPE`

Connections

Uses: `moe_pruning.data.calibration`, `moe_pruning.model.architecture`, `moe_pruning.model.instrumentation`, `moe_pruning.model.loader`, `moe_pruning.scoring.statistics`, `moe_pruning.utils.cli`, `moe_pruning.utils.paths`, `moe_pruning.utils.reproducibility`